Tarski's Circle-Squaring Problem: From Polygon Dissection to Borel Equidecomposition

N-20221214

§1.1 The route of the note

Tarski's circle-squaring problem unfolds through a sequence of increasingly subtle notions:

$$\text{Tarski circle-squaring} \longrightarrow \text{Hilbert's third problem}$$
$$\longrightarrow \text{measure and paradoxical decomposition}$$
$$\longrightarrow \text{Laczkovich's theorem} \longrightarrow \text{Borel circle squaring.}$$

The visual configurations below support theorem statements whose algebraic and measure-theoretic content is developed directly in the prose.

Tarski asked in 1925 whether a disk and a square of equal area in the plane are equidecomposable: can the disk be partitioned into finitely many pieces, and can those pieces be moved by isometries to partition the square? The link between discrepancy, bounded flows and definability follows Laczkovich and the Borel theorem of Marks and Unger.

§1.2 Equidecomposability and dissection congruence

Let a group G act on a set X. Two subsets $A, B \subseteq X$ are G-equidecomposable if there are finite partitions

$$A = A_1 \sqcup \cdots \sqcup A_m, \qquad B = B_1 \sqcup \cdots \sqcup B_m,$$

and elements $g_i \in G$ such that

$$g_i(A_i) = B_i.$$

For plane polygons, one usually lets G be the group of Euclidean isometries. In that setting the classical language is dissection congruence: cut one polygon into finitely many polygonal pieces and rearrange them to obtain the other.

The first example is the familiar parallelogram-to-rectangle cut.

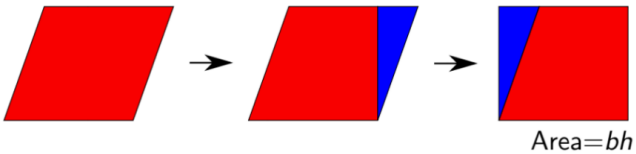


Figure 1.1: A parallelogram dissected and rearranged into a rectangle.

Area is obviously preserved by dissection congruence. The Wallace–Bolyai–Gerwien theorem says that, for plane polygons, area is also sufficient.

Theorem 1.2.1 (Wallace–Bolyai–Gerwien)
Any two plane polygons of the same area are dissection congruent.

One small point in the note is transitivity. If polygon P is dissection congruent to Q, and Q is dissection congruent to R, then P is dissection congruent to R. One refines the two dissections so that the intermediate polygon Q is cut compatibly, then composes the rearrangements.

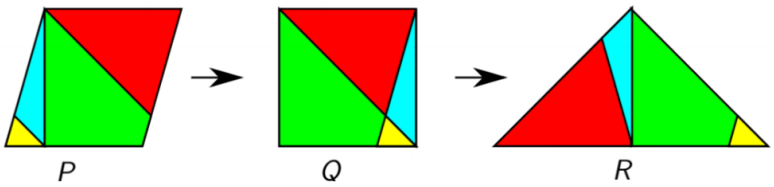


Figure 1.2: The transitivity intuition: a dissection from P to Q and one from Q to R compose.

The proof of the theorem is constructive. First chop a polygon into triangles.

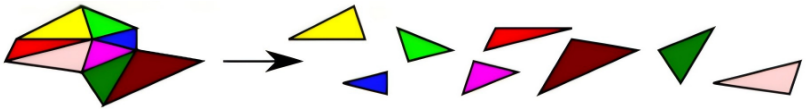


Figure 1.3: The first reduction: chop a polygon into triangles.

Then dissect each triangle into a parallelogram and then a rectangle; dissect each rectangle into a square; combine squares by the Pythagorean dissection.

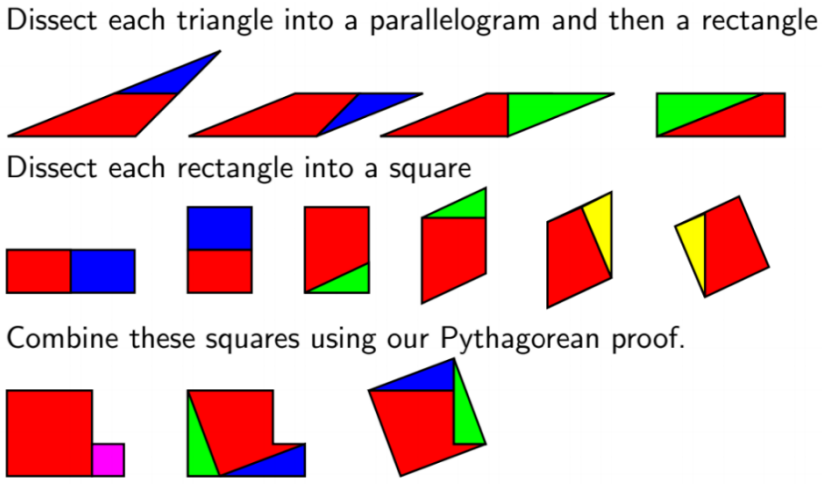


Figure 1.4: The visual chain: triangle to parallelogram/rectangle, rectangles to squares, then combine squares.

The theorem establishes the baseline: for tame plane polygons, equal area exactly characterizes finite dissection congruence.

§1.3 Hilbert's third problem and the Dehn invariant

Hilbert's third problem asks whether equal volume similarly characterizes dissection congruence of three-dimensional polytopes. Dehn's theorem says no: a cube and a regular tetrahedron of the same volume are not dissection congruent.

The extra obstruction is the Dehn invariant. If a polyhedron has edge lengths ℓ_i and corresponding dihedral angles θ_i, the invariant has the form

$$D(P) = \sum_i \ell_i \otimes \theta_i$$

with values in $\mathbb{R} \otimes_{\mathbb{Q}} (\mathbb{R}/\pi\mathbb{Q})$, where angles differing by rational multiples of π represent the same class. The invariant records length-angle data that volume does not see; the tensor product signals that this is no longer elementary area bookkeeping.

Sydler's theorem says that volume and Dehn invariant together are complete in dimension three. Thus the two-dimensional polygon theorem has no naive three-dimensional analogue.

§1.4 Measure-theoretic background

Measure introduces a fundamental obstruction. Vitali sets imply that for every $n \geq 1$, there is no extension of Lebesgue measure to all subsets of $\mathbb{R}^n$ that is both countably additive and invariant under isometries.

If countable additivity is weakened to finite additivity, the dimension matters. For $n \geq 3$, the Banach–Tarski paradox rules out a finitely additive isometry-invariant measure on all bounded sets compatible with volume, whereas Banach measures exist for $n \leq 2$. The reason is group-theoretic: in dimension at least three, the isometry group contains enough free-group behavior for paradoxical decompositions; in dimensions one and two it does not.

For Tarski's problem in the plane, this means that equal area is still a necessary condition. A disk and square that are equidecomposable by isometries must have the same Lebesgue measure, even if the pieces themselves are nonmeasurable.

§1.5 Tarski's circle-squaring problem

The central question is then:

> Are a disk and a square in $\mathbb{R}^2$ of the same area equidecomposable?

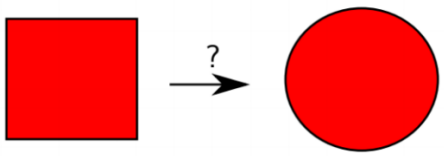


Figure 1.5: The central visual question: equal-area disk and square.

At first sight the boundary seems to obstruct the construction. A disk has circular boundary; a square has straight boundary. This intuition is correct for a stricter notion, but not for arbitrary equidecomposition.

§1.6 Scissors congruence and why nice pieces fail

The note distinguishes equidecomposability from scissors congruence. In the scissors-congruence version, the pieces must be tame: each piece is homeomorphic to a disk and has boundary given by curves of finite length. Under such restrictions, the disk and square are not scissors congruent.

The relevant signed boundary quantity is:

$$\text{convex circular perimeter} - \text{concave circular perimeter}.$$

When a curved circular arc is created inside a dissection, it is paired with an opposite concave arc, so the internal contributions cancel. The remaining outer boundary contribution is preserved. A disk has a nonzero circular-boundary contribution, while a square has none. Hence a disk and square cannot be scissors congruent in this tame sense.

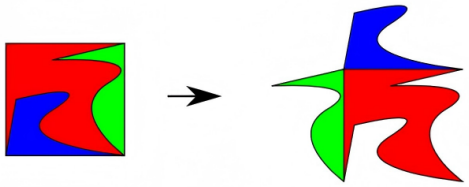


Figure 1.6: Tame scissors pieces preserve circular-boundary information.

This is the first major subtlety: the positive answer to Tarski's problem requires pieces much wilder than ordinary scissors pieces.

§1.7 Laczkovich's positive theorem

Laczkovich proved in 1990 that Tarski's circle-squaring problem has a positive answer. In 1992 he proved a broader theorem. A simplified statement is:

Theorem 1.7.1 (Laczkovich)
Let $A, B \subseteq \mathbb{R}^k$ be bounded sets with the same positive Lebesgue measure. If the boundaries of A and B have upper Minkowski dimension strictly less than k, then A and B are equidecomposable by translations.

For a disk and square in the plane, the boundaries are one-dimensional, so the hypothesis applies.

The proof does not draw a literal jigsaw. It converts the problem into a bounded matching problem in a graph generated by translations.

§1.8 First idea: work in the torus

Scale and translate A and B so that they lie inside $[0,1)^k$. Identify this cube with the k-torus

$$\mathbb{T}^k = (\mathbb{R}/\mathbb{Z})^k.$$

Then A and B are equidecomposable by translations as subsets of the torus iff, after possibly using more pieces, they are equidecomposable by translations in $\mathbb{R}^k$.

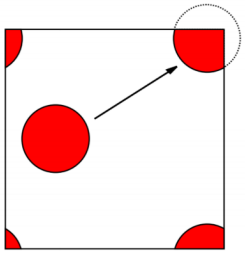


Figure 1.7: Working on the torus: translations wrap around the fundamental cube.

This torus reduction matters because translations become an action on a compact group. Compactness and periodic boundary conditions make the orbit structure manageable.

§1.9 Random translations and a free action

Choose a sufficiently large integer d and random translations

$$u_1, \dots, u_d \in \mathbb{T}^k.$$

They define an action of $\mathbb{Z}^d$ on $\mathbb{T}^k$ by

$$(n_1, \dots, n_d) \cdot x = n_1 u_1 + \cdots + n_d u_d + x.$$

For almost every choice of the u_i, this action is free: no nonzero element of $\mathbb{Z}^d$ fixes a point. Thus each orbit can be visualized as a copy of $\mathbb{Z}^d$.

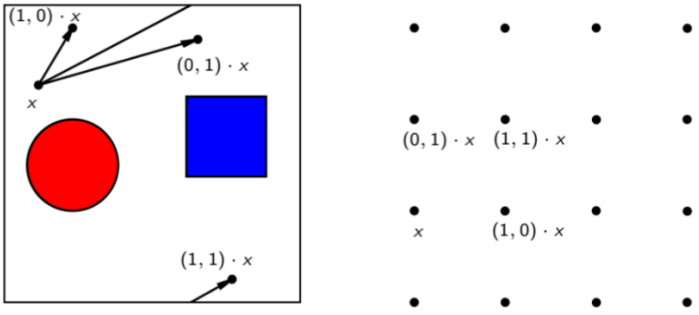


Figure 1.8: Random translations generate lattice-like orbits on the torus.

§1.10 The graph and bounded-distance bijection

Define a graph G with vertex set $\mathbb{T}^k$. Two points x, y are adjacent if there exists $g \in \mathbb{Z}^d$ with

$$g \cdot x = y, \qquad \|g\|_\infty = 1.$$

So edges correspond to single generator moves.

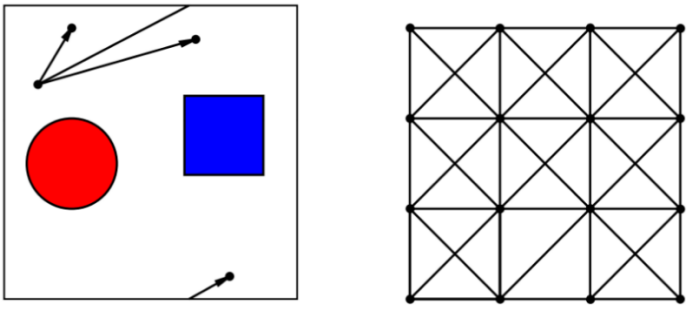


Figure 1.9: The generator graph: locally it looks like a lattice with bounded moves.

To show A and B are equidecomposable, it suffices to find a bijection

$$f : A \to B$$

of bounded distance in G: there is some fixed N such that

$$d_G(x, f(x)) \leq N \qquad \text{for all } x \in A.$$

If such an f exists, then only finitely many group elements g can occur. Put

$$A_g = \{x \in A : f(x) = g \cdot x\}.$$

The sets A_g partition A, and the translated sets $g \cdot A_g$ partition B. This gives the desired finite equidecomposition.

This is the conceptual heart of the proof:

$$\text{finite translation equidecomposition} \quad \Longleftarrow \quad \text{bounded-distance orbit matching.}$$

§1.11 Orbit matching

Inside a single orbit of the $\mathbb{Z}^d$ action, the problem becomes matching red A-points to blue B-points by bounded paths.

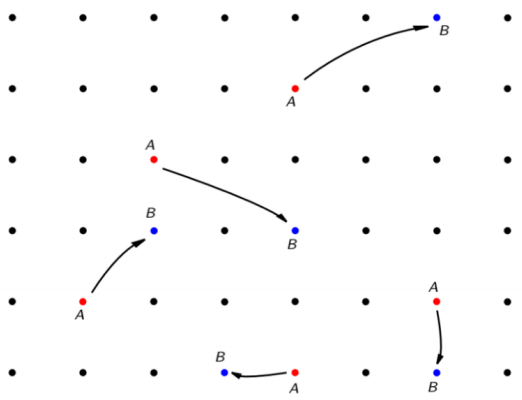


Figure 1.10: Within one orbit, equidecomposition becomes a matching problem.

A bounded matching can exist only if large finite regions of the orbit contain approximately the same number of A-points and B-points. This leads to discrepancy estimates.

§1.12 Discrepancy estimates

For an orbit point x, define a finite box

$$S_N(x) = \{(n_1, \dots, n_d) \cdot x : 0 \le n_i < N\}.$$

Ergodic intuition suggests

$$|S_N(x) \cap A| \approx \lambda(A) N^d.$$

This is only a heuristic: ordinary ergodic theorems alone are not strong enough. Laczkovich needs a uniform quantitative estimate from Diophantine approximation and discrepancy theory.

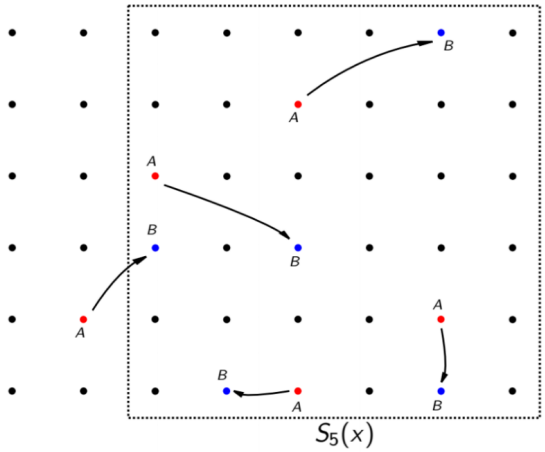


Figure 1.11: A finite orbit box $S_N(x)$; discrepancy compares observed counts with expected measure.

The key lemma states that there exist $\varepsilon > 0$ and M such that for all x and N,

$$\Big||S_N(x) \cap A| - \lambda(A) N^d\Big| \le M N^{d-1-\varepsilon},$$

and similarly for B:

$$\Big||S_N(x) \cap B| - \lambda(B) N^d\Big| \le M N^{d-1-\varepsilon}.$$

Since $\lambda(A) = \lambda(B)$, large boxes contain nearly equal numbers of A and B points. The error term grows more slowly than the volume term N^d, which is exactly what makes matching possible.

§1.13 From discrepancy to a bounded transport flow

The discrepancy estimate is useful because it places a quantitative restriction on how much mass can accumulate in one part of an orbit. Write

$$b(x) = \mathbf{1}_A(x) - \mathbf{1}_B(x)$$

for the surplus at a vertex of the orbit graph. A point of $A \setminus B$ contributes $+1$, a point of $B \setminus A$ contributes -1, and all other vertices have zero net surplus. A transport flow is an antisymmetric function on oriented edges,

$$\varphi(x, y) = -\varphi(y, x),$$

with discrete divergence

$$\boxed{\sum_{y\sim x}\varphi(x,y)=b(x).}$$

This is an exact conservation equation, not a statistical approximation.

Its meaning is already clear on four consecutive vertices v_0, v_1, v_2, v_3. Suppose $A = \{v_0, v_1\}$ and $B = \{v_2, v_3\}$. Sending one unit across v_0v_1, two across v_1v_2, and one across v_2v_3 gives

x	v_0	v_1	v_2	v_3
$b(x)$	1	1	-1	-1

because the outgoing minus incoming flows are exactly $1, 1, -1, -1$. The amount crossing each edge equals the cumulative imbalance on one side. The one-dimensional model explains the difficulty in higher dimensions: if a region has a large surplus, that surplus must be carried across the region's boundary.

Now use Laczkovich's discrepancy estimate for the same box $S_N(x)$, both for A and for B. Since $\lambda(A) = \lambda(B)$, subtraction gives

$$\left|\sum_{y\in S_N(x)} b(y)\right| \le 2MN^{d-1-\varepsilon}.$$

A d-dimensional box has $O(N^{d-1})$ boundary edges, so this surplus is smaller than the boundary scale by a factor $O(N^{-\varepsilon})$. At dyadic scales $N = 2^j$, the corresponding scale factors are summable:

$$\sum_{j\ge 0} 2^{-j\varepsilon} < \infty.$$

This is the critical quantitative feature. A mere ergodic estimate $o(N^d)$ would not control the amount of flow that must cross boundaries at successive scales. The stronger discrepancy estimate makes uniformly bounded multi-scale corrections plausible.

Constructing such a flow is a theorem-level step: the corrections must be routed consistently across overlapping boxes and across every orbit. The bound above explains the role of the exponent; it is not, by itself, a proof that the infinite network has a bounded solution. Laczkovich's method and its Borel refinement supply the additional combinatorial construction.

§1.14 Borel Pieces of a Real Flow

A uniformly bounded real-valued flow still does not immediately give a translation partition. A single point cannot be divided into a fraction of a point, whereas a real flow is permitted to carry noninteger quantities. An integer-valued flow is better adapted to matching discrete vertices.

In finite networks, one can see the difference by the integral max-flow theorem: when supplies and capacities are integers, a feasible flow can be chosen integer valued. For the infinite orbit graph there is an extra compatibility problem. Independently choosing an integral flow on each finite subgraph may fail to produce a globally bounded or definable choice.

There is also a second, logically separate issue. Even if every orbit admits an appropriate matching, making one arbitrary choice on each of uncountably many orbits need not yield a Borel function of the starting vertex. A valid equidecomposition by arbitrary pieces need not be a Borel equidecomposition.

The theorem of Marks and Unger solves both obstacles for bounded Borel sets $A, B \subseteq \mathbb{R}^k$ with

$$\lambda(A) = \lambda(B) > 0, \qquad \overline{\dim}_M(\partial A),\ \overline{\dim}_M(\partial B) < k.$$

Their proof first builds a bounded *Borel real flow* using discrepancy. It then uses special finite Borel partitions of free $\mathbb{Z}^d$-orbits, together with integral-flow arguments, to obtain a bounded *integer-valued Borel flow*. The final conversion produces a Borel bijection $f : A \to B$ that moves every point by one of finitely many translations. The underlying orbit partitions are a substantial input from Borel graph theory; bounded real flow alone is not the conclusion.

To see why this last property gives the desired pieces, let $F \subset \mathbb{Z}^d$ be the finite set of displacement vectors used by f. Define

$$A_g = \{x \in A : f(x) = g \cdot x\}, \qquad g \in F.$$

The sets A_g are Borel because f and the group action are Borel. They partition A, while the translates $g \cdot A_g$ partition B because f is a bijection. Only finitely many translations are required.

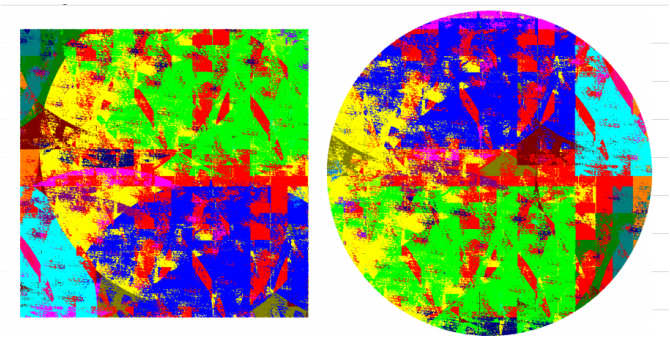


Figure 1.12: A schematic depiction of intricate Borel pieces: definability does not require polygonal or smooth boundaries.

For the circle and square, both boundaries have Minkowski dimension one, strictly below the ambient dimension two, so the theorem applies. The pieces are Borel, but they need not have finite-perimeter boundaries or resemble jigsaw pieces. A Borel decomposition is thus substantially more regular than a partition into arbitrary subsets, yet much less geometrically tame than ordinary scissors congruence.

One question has revealed three distinct invariance mechanisms. Euclidean polygon dissections are governed by area; polyhedral scissors congruence detects additional Dehn length-angle data; circle squaring is controlled by arithmetic discrepancy and graph transport once sufficiently complicated pieces are admitted. The regularity of those pieces determines which theorems can be used, and which obstructions survive.